

Dataset	classes	features only	eig (random)	eig _{best} (fair)	eig _{best} (oracle)	neural (fair)
MNIST-4	4	0.707	0.789	0.783	0.808	0.782
MNIST-10k	10	0.450	0.786	0.787	0.798	0.790
ALOI	100	0.737	0.773	0.767	0.787	0.795

Setup. Three clustering benchmarks, 4,000 samples each, scored by k -means AMI: MNIST-4 (digits 0–3, 4 classes), MNIST-10k (all digits, 10 classes), ALOI (100-object subset of the Amsterdam Library of Object Images). MDT embeds by plain truncated **svds** of its operator. Here k is the number of clusters (the known class count, which k -means also needs), and the embedding is the top- k singular vectors of the operator after dropping the trivial top mode (σ_1 , the constant stationary direction), scaled by their singular values; by Eckart–Young this is the optimal rank- k summary, and we keep exactly k because the clustering that follows asks for k groups (extra components only add noise on the near-uniform diffused operator). All columns share this decomposition: features only: k -means on raw features, no diffusion; eig (random): mean over 8 random trajectories; eig_{best} (fair): best of 8 selected by top- k spectral energy (unsupervised, deployable); eig_{best} (oracle): best of 8 selected by AMI (uses labels); neural (fair): a net reproducing the top- k SVD of the same energy-selected (fair) operator, by a Rayleigh objective (the Rayleigh trace $\text{tr}(Q^\top \mathbf{W} \mathbf{W}^\top Q)$ is a differentiable surrogate whose optimum equals the SVD, so gradient ascent learns the SVD as a function, whereas an eigensolver only computes it as numbers).

Result. With the decomposition fixed to **svds**, trajectory selection adds nothing (eig random $\approx$ eig_{best} fair); the oracle leads only by selecting on labels (AMI), so it is not deployable. Neural matches, and marginally exceeds, **svds** (ALOI +0.03), though within run-to-run noise: a hint, not yet a result, that a learned approximation may leave small headroom over MDT.

Table 2: Scaling on MNIST-10k (Scenario 2).

N	wall-clock time			accuracy	
	dense eigh	svds	neural	AMI (svds $\approx$ neural)	
2 000	4.3 s	0.07 s	18.5 s	0.74	
4 000	14.6 s	0.11 s	40.0 s	0.79	
8 000	98.0 s	0.30 s	71.3 s	0.81	
10 000	n/a	0.31 s	89.4 s	0.81	

Setup. Time to build the embedding of N points three ways, plus the resulting k -means AMI. dense **eigh** = classical $O(N^3)$ eigendecomposition of the dense operator; **svds** = truncated top- k SVD of the sparse, matrix-free operator action (Scenario 2); neural = the Rayleigh-trained net (Scenario 1). **svds** and neural produce the same embedding, hence one shared accuracy column; dense **eigh** is omitted at $N=10k$ (too slow).

Result. **svds** (0.07–0.31 s) is $\sim 300\times$ faster than dense **eigh** and $\sim 200\times$ faster than the net, at identical accuracy (0.74–0.81). The scalable classical method wins; the neural machinery buys no speed.

Table 3: Out-of-sample AMI over 5 seeds (Scenario 3).

Dataset	Nyström	deep ($\sqrt{\pi}$ -Gram)
MSRC-v5	0.685 $\pm$ 0.021	0.648 $\pm$ 0.093
Handwritten	0.859 $\pm$ 0.028	0.866 $\pm$ 0.030
Wikipedia	0.307 $\pm$ 0.102	0.295 $\pm$ 0.057
UCI	0.801 $\pm$ 0.054	0.826 $\pm$ 0.053
OutdoorScene	0.490 $\pm$ 0.041	0.480 $\pm$ 0.040
Caltech101-7	0.525 $\pm$ 0.044	0.527 $\pm$ 0.043

Setup. A held-out test split is embedded and clustered by two out-of-sample extensions of the same fixed operator $\mathbf{W}$ (mean $\pm$ std over 5 seeds; OutdoorScene 4, one seed dropped to a degenerate view). Nyström = the SVD’s own training-free extension: form the new point’s operator row $\mathbf{w}_*$ against the training set and project it onto the training singular vectors, $\psi(x_*) = \mathbf{w}_*^\top \mathbf{V}$. deep ($\sqrt{\pi}$ -Gram) = the DDM encoder f_θ (one MLP branch per view), the same encoder for every dataset, with no feature/affinity variants. It is trained by $\sqrt{\pi}$ -Gram matching: let $\boldsymbol{\pi}$ be the stationary distribution of $\mathbf{W}$ ($\boldsymbol{\pi}^\top \mathbf{W} = \boldsymbol{\pi}^\top$), $\mathbf{A} = \boldsymbol{\Pi}^{1/2} \mathbf{W} \boldsymbol{\Pi}^{-1/2}$ its $\sqrt{\pi}$ -symmetrised form, and $\mathbf{G} = \mathbf{A} \mathbf{A}^\top - \sqrt{\pi} \sqrt{\pi}^\top$ the target Gram (with the trivial top mode $\sqrt{\pi}$ removed). The loss makes the encoder’s $\sqrt{\pi}$ -scaled Gram match it, $\min_\theta \|(\sqrt{\pi} \odot F)(\sqrt{\pi} \odot F)^\top - \mathbf{G}\|_F^2$, $F = f_\theta(X)$. By Eckart–Young its optimum is the top- k eigenspace of $\mathbf{G}$ (the diffusion-map embedding), so the net learns that embedding as a function and extends to new points by a forward

pass. The $\sqrt{\pi}$ scaling is essential, not cosmetic: a constant (collapsed) embedding lands exactly on the removed $\sqrt{\pi}\sqrt{\pi}^\top$ mode, so it is penalised; drop it and the encoder collapses to a constant (verified here: $\text{AMI} \rightarrow 0.02$).

Result. Over 5 seeds Nyström and the encoder tie within one std on every dataset (row gaps ≤ 0.04 , inside the ± 0.02 – 0.10 spreads): out-of-sample is a wash. The single-run gaps that once looked decisive are seed noise. A faithfully-trained encoder does not beat the SVD’s free Nyström extension; at best it reproduces it.

Table 4: End-to-end learning of the geometry fails (Scenario 4).

Dataset	fixed+random	fixed+learned wts	learned geom.
MSRC-v5	0.682	0.470	0.328
Caltech101-7	0.535	0.410	0.049
OutdoorScene	0.570	0.463	0.005
UCI	0.822	0.914	0.751

Setup. AMI as progressively more of the operator is made trainable (left→right). fixed+random: the per-view kernels P_v (Gaussian kNN affinities) and the fusion are both fixed, a random convex trajectory, no learning. fixed+learned wts: keep the kernels P_v fixed and learn only the mixing weights, the $t \times V$ matrix a (one softmax over views per step) that convexly combines the fixed per-view walks, $\mathbf{W} = \prod_s (\sum_v a_{sv} P_v)$, by Adam on the contrastive loss (positive pairs = kernel neighbours). learned geom.: also learn the geometry itself: a per-view feature map ϕ_v defines each kernel $P_v(\phi_v)$, so both the kernels and a are trained end-to-end under the same loss (now even the Gram target moves). Four datasets; bold = near-zero collapse.

Result. Accuracy falls monotonically as more is learned: fixed > learned-weights > fully-learned (near-zero on Caltech/OutdoorScene). Learning the operator hurts. The mechanism is optimisation/conditioning (flat loss, spectrum flattening), not the literal rank-one collapse the numbers suggest.

Table 5: t -sweep of three methods on the four collapse datasets (AMI).

Dataset	method	$t=1$	$t=2$	$t=3$	$t=4$
MSRC-v5	fixed MDT	.704	.736	.728	.720
	imit. (Gram)	.712	.724	.701	.706
	end-to-end	.533	.309	.277	.239
Caltech101-7	fixed MDT	.547	.588	.619	.620
	imit. (Gram)	.643	.545	.647	.644
	end-to-end	.234	.094	.070	.085
OutdoorScene	fixed MDT	.629	.633	.606	.598
	imit. (Gram)	.579	.561	.522	.525
	end-to-end	.340	.001	.098	.177
UCI	fixed MDT	.846	.834	.830	.840
	imit. (Gram)	.841	.810	.840	.828
	end-to-end	.867	.837	.763	.730

Setup. For diffusion length $t=1..4$, three ways to get the embedding. fixed MDT = best-of-8 length- t operator, plain SVD (no learning); imit. (Gram) = a $\sqrt{\pi}$ -Gram encoder trained to reproduce that fixed operator; end-to-end = learn the operator under the contrastive loss (Scenario 4, 5-seed mean). Four datasets.

Result. Fixed and imit are stable across t and coincide (imit reproduces the fixed operator at every length), while end-to-end collapses as t grows (well-connected kNN graphs over-mix past one hop), except UCI (one dominant view).

Why imit beats end-to-end. Two reasons. (i) Fixed vs. moving target: imit matches the Gram of a fixed, known-good operator (the best-of-8 MDT), so its target never drifts; end-to-end instead learns the operator, so its target $\mathbf{G}(\theta)$ moves and slides into a degenerate, spectrum-flattened optimum. (ii) Conditioning: Gram matching is a well-behaved quadratic, $\|FF^\top - \mathbf{G}\|^2$ with gradient $4(FF^\top - \mathbf{G})F$, so it descends cleanly, whereas the contrastive loss saturates once the operator’s rows are near-uniform (its exp/softmax gives tiny, near-equal gradients) and stalls. imit therefore reaches higher AMI because it inherits the good fixed operator intact, while end-to-end degrades that operator as it trains.

Table 6: Unsupervised MDT trajectory selection (Scenario 4).

Dataset	Spearman ρ			selection regret		
	Q_c	sil	en	Q_c	sil	en
MSRC-v5	−.25	−.17	.49	.072	.097	.043
Yale	−.12	−.59	.11	.073	.131	.064
BBCSport	.13	−.04	.60	.125	.164	.019
Caltech101-7	−.12	.11	.36	.047	.467	.043
Handwritten	.24	.05	.19	.050	.365	.209
UCI	−.28	−.30	.36	.369	.404	.121
Wikipedia-test	.33	.61	−.47	.000	.000	.315
mean	−.01	−.05	.23	.105	.233	.116

Setup. Can an unsupervised index pick the best trajectory? Over 30 random trajectories $\times 3$ seeds on structured datasets (oracle AMI ≥ 0.35), rank trajectories by three indices and score each two ways: Spearman ρ with the oracle-AMI ranking (higher better) and selection regret = oracle–selected AMI (lower better). Indices: Q_c = MDT’s contrastive Internal Quality, sil = embedding silhouette, en = spectral energy (top- k explained variance). Embedding dim $d=k$; bold = best per block.

Result. Spectral energy is the only positive ranker (mean $\rho +.23$ vs $-.01, -.05$) and lowest-regret on 5/7, but it is weak and fails on heterogeneous-view Wikipedia-test ($\rho -.47$; four more cross-modal cases below). Raising d (64/128) inflates ρ /regret yet lowers deployed AMI ($0.54 \rightarrow 0.52$), a flat-spectrum mirage, not a gain. No index selects reliably.

Why energy fails on heterogeneous views. Energy ranks trajectories by spectral cleanliness, not cluster correctness. On cross-modal data a view can be internally clean (high energy) yet uninformative about the labels (low single-view AMI), a clean-but-useless view; a trajectory that leans on it keeps high energy but clusters poorly, so energy anti-correlates with AMI. A separate probe of five heterogeneous datasets (per-view energy vs. single-view AMI, and selector ρ over 25 trajectories $\times 3$ seeds) shows energy fails exactly when such a view is present:

Dataset	views	energy ρ	clean-but-useless view (en/AMI)	verdict
Cora	4	−.61	v2, v3: .14/.00	fails
Wikipedia	2	−.59	v0: .07/.04	fails
Prokaryotic	3	−.38	v2: .11/.04	fails
NUS-WIDE	5	−.34	v3: .06/.04	fails
CiteSeer	2	+.93	none	works

Energy fails ($\rho < 0$) on the four datasets carrying a clean-but-useless view and works ($\rho +.93$) on the one without. Cora is the sharpest: its two highest-energy views (.14) have AMI $\approx .00$, while the most informative view (AMI .13) has the lowest energy (.01), so energy points the wrong way (energy-vs-AMI across views $\rho = -.95$). Energy measures clean structure, not correct structure, so it cannot detect a clean-but-useless view: the failure is fundamental, not a tuning issue.